Whale Watching and the Disturbance Behind It: Evidence, Stakeholders and Management in Australian Waters

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1. The social change issue

Whale watching lets people get close to animals most of us would otherwise never see, but that closeness is also what causes problems. Boats can disturb whales through underwater noise, changes in behaviour and stress (Sprogis et al., 2020; Lemos et al., 2022), and the International Whaling Commission has been pushing whale-watching principles since 1996 (International Whaling Commission, n.d.-a). So the question we're asking is how whale-watching tourism can be managed so people can still experience whales, along with the jobs, tourism and education that depend on them, without putting unacceptable biological costs on the whales themselves.

Scope. After comparing our individual reports, we agreed to focus on whale watching in Australian waters, with Jervis Bay, NSW, as our main case and international studies filling in where Australian data was thin. We each took a different angle: biological cost, Jervis Bay marketing, and governance and compliance.

Why this is social change. Reducing disturbance takes more than new science. Operators have to change how they work, marketing has to change, and tourists have to expect something different. Humpbacks pass the NSW coast between May and November (NSW Government, n.d.), which fills a winter gap for regional businesses and gets promoted through campaigns like Go Grand (Campaign Brief, 2025). So the whales that need protecting are the same ones the local economy relies on.

Whale watching also has real benefits worth protecting. It can teach tourists about whale behaviour and how to avoid disturbing them (International Fund for Animal Welfare [IFAW], 2025). Cárdenas et al. (2021) found most of the tourists they surveyed backed whale conservation after a tour. So the question isn't really whether whale watching is good or bad. It's when its benefits outweigh its costs, and who gets to decide that.

Rules. Table 1 sums up the NSW requirements, which get stricter when a calf is around. Some places go further. Canada, for example, requires vessels to stay 1,000 m away from endangered Southern Resident killer whales (Fisheries and Oceans Canada, 2026).

Table 1: NSW approach rules for whales (NSW Government, n.d.)

Who	Requirement
Vessels, including kayaks and surfboards	At least 100 m; at least 300 m if a calf is present
Swimmers and divers	At least 30 m
Approach	At an angle from the side, never from behind or ahead of a mother and calf; maximum three vessels in the caution zone

2. Public discourse and media

We pulled together ten items from our individual research and coded each one by source type, main frame, audience and the evidence it used (Table 2). We picked these by hand rather than systematically, so they show the range of framing, not how common each frame is.

Table 2: Coding of public communication items

Item	Frame	Audience	Evidence used
CP24 (2026)	Viral incident: vessel near orcas	General public	Video; agency assessment
Fisheries and Oceans Canada (2026)	Regulation: 1,000 m rule	Boaters, operators	Legal distance
Whale and Dolphin Conservation (2019)	NGO advocacy	General public	Everyday analogies
Mongabay (2026)	Technological solution	Conservation readers	Acoustic monitoring
NSW Government (n.d.)	Regulation and safety	Boaters	Legal distances
Campaign Brief (2025)	Regional winter campaign	Sydney and Canberra visitors	None stated
Woolage (2025)	Local news: whale census	Residents, visitors	Citizen counts
Jervis Bay Wild (n.d.)	Operator marketing	Tourists	Experience, reviews

Item	Frame	Audience	Evidence used
Shoalhaven Tourism (n.d.)	Destination guide	Visitors	None stated
Huntsdale (2026)	Cultural and educational tour	General public	Cultural knowledge

Three things stood out. First, the framing changes depending on who it's aimed at. Regulatory and NGO items talk about risk to boaters and the public, while promotional items talk to would-be visitors about convenience and experience. Second, the invisible harms mostly go unmentioned. Only the NGO campaign, which describes interrupted feeding, resting and nursing, and the acoustic monitoring story deal with them directly, while the most shareable item is footage of a close approach (CP24, 2026). Third, the evidence is mostly implied. Regulators cite rules, NGOs use everyday analogies and marketing leans on experience. So the people who drive demand, tourists, are the ones least likely to hear the evidence-based messages.

This matters because the way an issue is talked about decides which problem people see. If disturbance is understood as a rule-breaking close approach, following the distance rules seems to fix it, and quieter exposure gets missed. The technology frame (Mongabay, 2026) could also reassure people that noise is already being dealt with. Our regulatory items (Fisheries and Oceans Canada, 2026; NSW Government, n.d.) say what's required but not why, which probably limits their pull with tourists. Coverage of an Indigenous-led tour (Huntsdale, 2026) goes a different way, selling the experience on cultural meaning rather than getting close. We included Canadian items because they show the frames clearly, but Australian audiences might react differently.

3. Relevant evidence

Table 3 sets out each study's design, finding and main limitation, including causal strength and fit to Australian conditions.

Table 3: Key evidence and its limits

Study	Setting and design	Finding	Main limitation
Rolland et al. (2012)	North Atlantic right whales; natural experiment after reduced shipping	Lower low-frequency noise, lower stress hormones	Observational; one population

Study	Setting and design	Finding	Main limitation
Lemos et al. (2022)	Gray whales, Oregon; three field seasons	Previous-day vessel traffic linked to stress hormones, controlling for sex, age, reproductive state	Correlational
Sprogis et al. (2020)	Humpbacks, Western Australia; noise playback to resting mother-calf pairs	Louder noise: about 30% less resting, doubled respiration, 37% faster swimming	Short-term; playback, not real tours
Arranz et al. (2021)	Three whale-watching boats, Australia and Canary Islands	Noise reaching whales differed by boat type at the same speed and distance	Three boats
New et al. (2015)	Review of impact assessment	Boat disturbance significant; long-term population effects unclear	Evidence gaps
Scuderi et al. (2023)	Strait of Gibraltar; 637 questionnaires and observation	Operators only partly followed rules	Other jurisdiction
Affatati et al. (2024)	Operator knowledge and compliance	Good distance compliance, weak awareness of wider impacts	Limited generalisability
Cárdenas et al. (2021)	Tourists, Panama; survey	About 75% endorsed whale conservation after a tour	Self-reported attitudes
León et al. (2025)	Online reviews	Satisfaction fell when too many boats shared a location	Reviews are self-selected

What the biological evidence supports. The physiological and behavioural studies all point the same way, across three species, different regions and different designs. Sprogis et al. (2020) is the closest link to whale watching, because noise alone, at controlled levels, changed how much the whales rested, breathed and swam. Arranz et al. (2021) adds that noise depends on the type of boat, so a distance rule on its own can't control exposure. The catch is that most studies are short-term or correlational, and New et al. (2015) conclude that long-term effects on populations still aren't proven. So we treat disturbance as established and harm to populations as uncertain.

Why convergence matters. The designs differ (an unplanned drop in shipping, controlled field data, and direct noise manipulation), so their agreement counts for more. Still, hormone levels

and resting time are stand-ins. They show that animals respond, not what that response costs them over a lifetime.

Economic evidence. NAB merchant data reported by Destination NSW (2024) show Jervis Bay spending jumping by 174% for accommodation, 300% for restaurants and bars, and 239% for other retail, and roughly 450,000 people visit each year (Australian Government Department of Infrastructure, Transport, Regional Development, Communications and the Arts, n.d.). The figures as reported don't give a comparison period or separate out whale watching, so they tell us how much the region leans on tourism, not how much of that is whale watching.

Source quality. The biological studies are peer-reviewed, so we trust them most. Operator and tourism pages show how whale watching is sold, not what it does to whales, and CP24 and Mongabay report on things rather than test them. The governance and attitude studies (Scuderi et al., 2023; Cárdenas et al., 2021; León et al., 2025) tell us what people say, not what whales experience.

How stakeholders use evidence. In the items we coded, regulators lean on rules grounded in science, NGOs use scientific findings through analogies, and tourism promotion mostly doesn't mention disturbance evidence at all. We read that as choosing to emphasise the experience, not as denying the evidence.

Definitive statements.

- *Sound:* vessels and their noise cause measurable short-term behavioural and stress responses in several whale species.
- *Unclear:* whether those responses reduce reproduction or survival across a whole population.
- *Unsupported:* the claim that operators routinely and deliberately harass whales. Scuderi et al. (2023) found partial compliance, while Affatati et al. (2024) found good compliance on distance but weak understanding of wider impacts. Knowledge gaps and commercial incentives explain that better than bad intent.

What the evidence can currently tell us. It supports being cautious about noise and vessel numbers, but it can't yet tell us how many boats per whale is safe. That's why monitoring and review are part of our recommendations.

4. Major stakeholders

Table 4 maps stakeholders by their position and how much they influence whale exposure, based on the roles they play in the evidence and discourse above. Having influence isn't the same as having a voice.

Table 4: Stakeholder map

Stakeholder	Position and motivation	Influence on whales	Voice in debate
Whales	Undisturbed feeding, resting, nursing	Directly affected	None
Operators	Profit, reviews, workable rules	High: they set approach behaviour	Medium
Tourists and influencers	Close, shareable encounters; ethical expectations	Medium: demand and social media	High (online)
Regional tourism bodies and businesses	Winter income, jobs, occupancy	Medium: they promote the season	High
Government and regulators (NSW NPWS, Commonwealth)	Protection balanced with economy	High: rules and enforcement	Medium
Scientists	Measure effects, acknowledge uncertainty	Low directly; high via evidence	Medium
NGOs	Precaution, public pressure	Low directly	High
Indigenous communities	Cultural interests; views get little attention in commercial-tour debates	Low to medium: could help manage disturbance beyond licensed operators	Low

Agreement and disagreement. Everyone broadly agrees that whales should be protected and that the industry needs healthy populations. They disagree on how strict the rules should be and who pays for them. Dimmock et al. (2014) found that Australian managers put species health first, while operators wanted clearer guidance on compliance, and Finkler and Higham (2020) report little collaboration between operators, NGOs and scientists. The interaction is lopsided too: promotional actors rarely engage with the science, and scientists rarely reach tourists. Whales are the most affected stakeholder but get no say, while governments and operators strongly shape rules and practice, so decisions don't always reflect what whales need.

Possible biases. Operators and tourism bodies gain from close, reliable encounters. NGOs present findings through a conservation lens, scientists work within the limits of their data, and regulators have to write enforceable rules without full certainty. Our own group came with different emphases too, which we come back to in Section 8.

Who is missing? Indigenous communities are the least visible stakeholder. Commercial-tour debates give their views little attention, yet including them could address cultural interests and help manage disturbance beyond licensed operators. They only show up in one of our sources (Huntsdale, 2026), and we didn't engage with them directly. Responsibility is also split across levels of government (Australian Government, 2021), which makes it harder to say who is accountable for monitoring and enforcement.

5. Synthesis and discussion

The central challenge is running an industry that delivers social and economic benefits without disturbing the whales. Putting our three angles together shows five links that none of them shows on its own.

1. The visibility gap shapes the rules. Close approaches are easy to see and share, so they dominate the conversation, and they're what distance rules regulate. But the strongest evidence is about noise, which you can't see (Sprogis et al., 2020; Arranz et al., 2021). Distance matters but isn't enough. Speed, engine noise, time spent near whales and the number of vessels count too. NSW's three-vessel limit and approach angles already go beyond distance, but we found no equivalent standard for noise in the rules we looked at.

2. Marketing links demand to pressure. Promoting Jervis Bay whale watching as environmentally friendly and effortless (Campaign Brief, 2025; Jervis Bay Wild, n.d.) may raise expectations of close encounters, and operators depend on reviews. That adds to Finkler and Higham's (2020) point. The communication gap isn't only between operators, NGOs and scientists. It's also between conservation evidence and the audience that creates demand. León et al. (2025) suggest tourists dislike crowding too, which could be a useful lever.

3. Rules exist, but compliance and understanding are partial. Distance compliance is good in some places and partial in others, and wider impacts are poorly understood (Affatati et al., 2024; Scuderi et al., 2023). That points to monitoring as well as rule-setting, and to ongoing operator training that explains the biological reasons behind the rules (Affatati et al., 2024). It also fits Dimmock et al.'s (2014) finding that operators want clear guidance.

4. Regional dependence limits regulation. Jervis Bay's off-season income means strict limits have real local costs. So governing this is a balancing act under uncertainty, which favours adaptive management, where rules get reviewed as evidence comes in (International Whaling Commission, n.d.-b). Weak regulation could also undermine tourism's long-term future, so rules need to rest on evidence.

5. The industry relies on what it may disturb. Operators and conservation groups have different priorities but both want whales to keep showing up, yet the effects on populations are the least certain part of the evidence. That's the central tension we found: decisions need simple answers, and the science can only give qualified ones.

Relation to the literature. Our findings agree with Finkler and Higham (2020) and Dimmock et al. (2014) that evidence alone doesn't change practice. We add that the visibility gap and promotion shape what the public sees, and that being confident about harm to individual whales isn't the same as being confident about populations.

So change needs existing evidence translated into forms each stakeholder can use: enforceable rules, operator training and tourist education, delivered through channels tourists already notice, like tour operators and Go Grand.

6. Group position and recommendations

Position. We don't think whale watching should be rejected, but it should be managed with evidence-based limits on proximity, noise and interaction, reviewed regularly as the evidence improves. Where the benefits and whale welfare clash, whale welfare should come first, though the benefits are still worth keeping. We weighed a ban (not supported, since documented effects are short-term, effects on populations are unclear and regional communities would bear the cost), the status quo of distance rules (which ignores noise and relies on partial compliance) and adaptive management, which lets rules tighten or relax as evidence improves and fits both the strength and the limits of what we know.

Table 5: Recommendations

Who	Should do what	Why and how it helps
Regulators	Add vessel-noise and speed standards, test quiet-vessel criteria, monitor compliance and review rules on a set cycle	Noise varies by boat (Arranz et al., 2021); uncertainty requires adaptive review
Operators	Run a briefing before each tour on noise and distance; train crews; avoid guaranteeing close encounters	Closes the knowledge gap (Affatati et al., 2024) and reduces expectation pressure
Regional tourism bodies	Promote responsible viewing and land-based watching from lookouts at peak times, rather than adding boat trips	Maintains winter income with lower disturbance and less crowding (León et al., 2025)
Scientists and NGOs	Study long-term population effects; explain invisible harm in plain terms	Fills the key evidence gap and the visibility gap

Who	Should do what	Why and how it helps
Tourists and influencers	Choose operators who follow rules; avoid content rewarding close approaches	Demand shapes operator behaviour
Government as convener	Convene scientists, operators, local communities and Indigenous groups in setting and reviewing rules	Participation keeps rules workable and supports adaptive management (International Whaling Commission, n.d.-b)

Facilitators and barriers. A few things work in our favour: a shared interest in healthy whale populations, existing national guidelines, and tourists' own dislike of crowding. The barriers are economic dependence, the cost of enforcement, gaps in the evidence, promotion that rewards closeness, and the gap between invisible harm and visible rules.

Measuring success. Track compliance audits, vessel-noise measurements, operator training completion and repeat monitoring of mothers and calves. Adjust the rules if the indicators get worse, and report the results publicly.

7. Limitations and lessons learned

We relied on published studies and a small, hand-picked media sample. We collected no primary data, and some of our Australian evidence came from other countries. Economic data specific to whale watching was limited, so we used spending across all activities, which can't separate out whale watching's own contribution. One study (Sprogis et al., 2020) also carries a lot of our argument, and our stakeholder ratings are our own judgements, not measurements. If we did this again, we'd agree on scope earlier, sample media systematically with a coding frame, and talk to an operator or regulator.

8. Group reflection

Our individual reports agreed that adaptive, evidence-informed management is needed, but they leaned in different directions. One was more critical of operators and marketing and favoured alternatives like land-based watching, while the others saw operators as partly on conservation's side, citing Affatati et al.'s (2024) compliance findings. We sorted this out by weighing the evidence: partial compliance is documented, but deliberate harassment isn't. Nobody changed their position. The differences were in emphasis, not direction.

Feedback on our individual reports was that, even though we'd collected a good amount of evidence, we didn't go into enough detail about it. Here we responded by laying evidence out by design, finding and limitation (Table 3). We also removed AI-simulated stakeholder ratings because they weren't real evidence.

We all enjoy whale watching, but it's always been a moral dilemma, and we tend to overthink going to events like these. That shows in our position: we didn't want to reject whale watching, but we couldn't treat its benefits as equal to whale welfare.

Working on our own first kept our perspectives different, and we didn't run into any major problems working together, but it left us with overlapping sources and uneven quality. Next time we'd plan one clear path for the evidence from the start and stick to our focus, Australian waters and Jervis Bay, using international evidence only where Australian data was thin.

9. AI use acknowledgement

AI (gemini) was used to check for broken links, referencing mismatches, and staying within the word limit, including suggesting where to shorten content. As for individual reports from where information was taken, respective AI use remains valid.
https://share.gemini.google/I8rlkPpVL4FI

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